

VentureAI

AI Investment Committee — Project Spec

Band of Agents Hackathon · lablab.ai · June 12–19, 2026

Project Overview

An AI-powered venture capital investment committee that evaluates any startup using multiple specialized agents debating through Band, producing a Goldman Sachs-style investment memo in minutes.

Track	Track 3 — Regulated & High-Stakes Workflows
Prize Pool	\$10,000+ (1st: \$3,500 · 2nd: \$2,500 · 3rd: \$1,500)
Stack	Python · FastAPI · React · 21st.dev · Band · Featherless · AI/ML API
Submission	June 19, 2026

Problem

Early-stage VC screening is slow, expensive, and inconsistent. Associates spend days manually researching startups, and smaller funds can't afford the analyst talent that top firms have. A single analyst's bias can kill a good deal or pass on a bad one.

Solution

VentureAI deploys 5 specialized agents through Band that independently research different dimensions of a startup, debate findings, and produce a structured investment recommendation — the same workflow a real investment committee follows, compressed from days to minutes.

How It Works

- User inputs a startup — company name, website URL, or uploaded pitch deck PDF
- Sourcing Agent processes the input and publishes a structured startup profile to Band
- Market Research, Founder Research, and Financial agents run in parallel through Band
- Devil's Advocate Agent receives all three outputs and challenges every positive finding
- Investment Committee Agent synthesizes the full debate into a final investment memo
- Human reviews the recommendation before it's finalized (human-in-the-loop checkpoint)

Agent Communication Flow (Band)



The Agents

● 1. Sourcing Agent

Model	Llama 3.1 70B via Featherless
Role	Entry point and intake processor

What it does:

Takes the raw user input — company name, URL, or PDF — and extracts structured data about the startup. Kicks everything off and packages information for all downstream agents.

Inputs:

- Company name / URL / pitch deck PDF (processed via AI/ML API)

Outputs to Band:

- company_name, one_liner, stage, industry
- founders list, funding history, location, website

- raw_description for downstream agents

Band behavior:

Publishes startup profile to Band room. Notifies Market Research, Founder Research, and Financial agents simultaneously to begin parallel research.

● 2. Market Research Agent

Model	Mistral 7B via Featherless
Role	Market and competitive landscape analyst

What it does:

Researches the market the startup is operating in. Evaluates whether the market is real, large enough, and whether the timing is right. Also maps the competitive landscape and identifies tailwinds or headwinds.

Inputs:

- Sourcing Agent's startup profile (from Band)

Outputs to Band:

- TAM / SAM / SOM estimates
- Competitor list and differentiation analysis
- Market growth trajectory and timing verdict
- Regulatory tailwinds or headwinds
- Market score (0-100) + key risks

Band behavior:

Publishes market analysis to Band. Devil's Advocate Agent picks this up alongside the other two research outputs.

● 3. Founder Research Agent

Model	Qwen 2.5 72B via Featherless
Role	Team and founder evaluator

What it does:

Investigates the people behind the company. At early stage, team quality is the #1 factor VCs evaluate. Looks for domain expertise, prior exits, red flags, and whether the team is uniquely positioned to win.

Inputs:

- Sourcing Agent's startup profile — specifically founders list (from Band)

Outputs to Band:

- Per-founder background, domain expertise, red flags
- Team completeness assessment (tech + business + domain)
- Prior exit history
- Founder score (0-100) + summary

Band behavior:

Publishes founder analysis to Band. Runs in parallel with Market Research and Financial agents.

● 4. Financial Agent

Model	Mistral 7B via Featherless
Role	Business model and unit economics evaluator

What it does:

Evaluates whether the business model makes financial sense. Looks at revenue model, unit economics, burn rate vs raise amount, and path to profitability. Based on pitch deck or publicly available info — a credibility check, not a full audit.

Inputs:

- Sourcing Agent's startup profile (from Band)

Outputs to Band:

- Revenue model identification and assessment
- Unit economics (CAC, LTV, margins) if available
- Raise amount vs burn rate reasonableness
- Path to profitability assessment
- Financial score (0-100) + red flags

Band behavior:

Publishes financial analysis to Band. Runs in parallel with Market Research and Founder Research agents.

● 5. Devil's Advocate Agent

Model	DeepSeek R1 via Featherless
Role	Contrarian challenger — the most important agent

What it does:

Receives ALL three research agents' outputs and actively tries to tear them apart. Job is to steelman the bear case, surface risks nobody mentioned, and challenge every positive finding. Uses DeepSeek R1 specifically because it needs to reason carefully through counterarguments. This is what separates VentureAI from a simple pipeline.

Inputs:

- Market Research Agent output (from Band)
- Founder Research Agent output (from Band)
- Financial Agent output (from Band)

Outputs to Band:

- Challenges to every positive market finding
- Attacks on founder credentials
- Holes in the financial model
- Risks the other agents missed

- Top 3-5 failure modes for this startup
- Bear case score (0-100)

Band behavior:

Publishes bear case to Band. Investment Committee Agent waits for this before synthesizing the final verdict.

● 6. Investment Committee Agent

Model	Llama 3.1 70B via Featherless
Role	Final decision maker and memo writer

What it does:

Receives the full context from all agents through Band — bull case from research agents and bear case from Devil's Advocate — and synthesizes everything into a final investment recommendation with a full memo.

Inputs:

- Market Research Agent output (from Band)
- Founder Research Agent output (from Band)
- Financial Agent output (from Band)
- Devil's Advocate Agent output (from Band)

Outputs to Band:

- PASS / INVEST / WATCH verdict with confidence score
- Scores across market, team, financials, risk
- Executive summary
- Full investment memo
- Due diligence questions if recommending INVEST or WATCH
- Suggested valuation range

Band behavior:

Publishes final memo to Band. Triggers human review checkpoint in the UI before the memo is finalized.

Sample Investment Memo Output

VERDICT: INVEST ■ Confidence: 74%

EXECUTIVE SUMMARY

Airbnb is a two-sided marketplace for short-term home rentals...

MARKET OPPORTUNITY Score: 85/100

TAM \$100B+, 32% YoY growth, right-on-time entry...

TEAM ASSESSMENT Score: 78/100

Strong design + business split, no prior exits but high conviction...

FINANCIAL ASSESSMENT Score: 80/100

Asset-light model, favorable unit economics, \$600K raise reasonable...

BEAR CASE (Devil's Advocate) Score: 62/100

Regulatory risk in top cities, trust/safety incidents, hotel lobbying...

OVERALL SCORE: 76/100

DUE DILIGENCE QUESTIONS

- 1. How are you managing regulatory risk in NYC and SF?
- 2. What is your current host acquisition CAC?
- 3. How do you handle trust and safety incidents at scale?

SUGGESTED VALUATION RANGE

\$4M - \$6M pre-money based on comparable seed-stage marketplaces

Technology Stack

Band	Agent coordination, context sharing, task handoffs, parallel execution. Use promo code BANDHACK26 for free Pro month.
Featherless	LLM inference for all agents — Llama, Mistral, Qwen, DeepSeek. Use promo code BOA26 for \$25 credits.
AI/ML API	PDF pitch deck processing and multimodal input handling. Claim \$10 credits at lablab.ai/redeem-coupon/ai-ml-api-coupon-band-hackathon.
FastAPI	Backend server, REST endpoints, Band webhook handling.
React + 21st.dev	Frontend UI with live agent activity feed and investment memo display.
Python	All agent logic and backend.

7-Day Build Plan

Day	Focus	Tasks
Day 1	Setup	Claim all promo codes · Set up Band agents · Scaffold FastAPI + React · Get one agent talking to Band
Day 2	Core agents	Build Sourcing Agent + Market Research Agent · Test Band communication · Use Airbnb early pitch as test case
Day 3	Remaining agents	Build Founder Research + Financial Agent · Test full parallel execution through Band

Day 4	Devil's Advocate + Committee	Build Devil's Advocate + Investment Committee · Connect backend to frontend
Day 5	Polish	Live agent activity feed in UI · PDF upload via AI/ML API · Test with 3-4 real startups
Day 6	Demo prep	Record demo video · Write README · Build slide deck for presentation
Day 7	Buffer + Submit	Fix anything broken · Final tests · Submit before June 19 deadline

Key Differentiators vs Other Submissions

- ✓ Parallel agent execution — not a linear pipeline
- ✓ Devil's Advocate creates genuine non-linear debate loop through Band
- ✓ Real investment memo output — not just a summary or approve/reject stamp
- ✓ Human-in-the-loop checkpoint before finalizing
- ✓ Works with pitch deck PDF, URL, or just a company name
- ✓ Live agent activity feed during processing — judges watch agents argue in real time
- ✓ 6 specialized agents vs the minimum 3 required

Good luck — let's build something sick ■